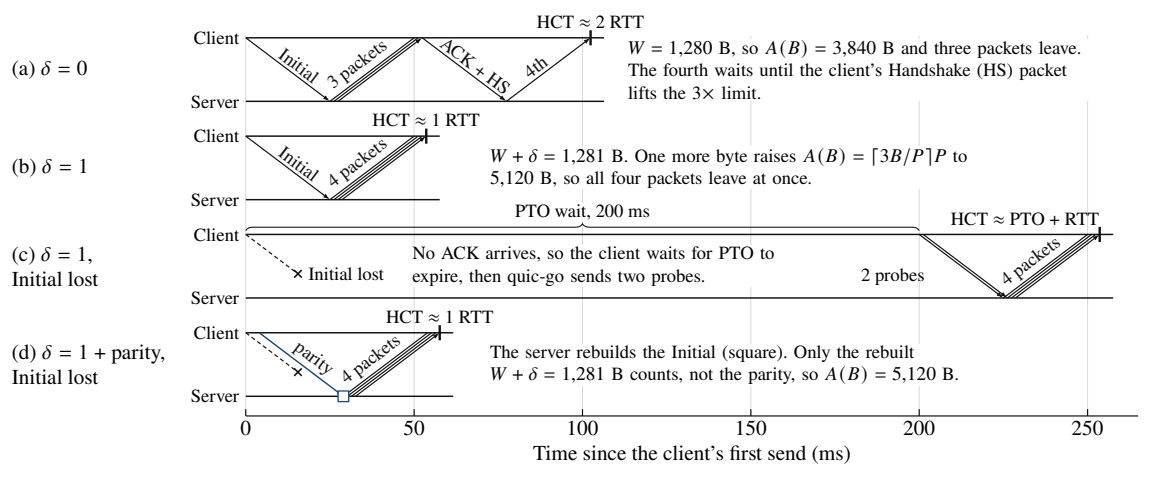

(a) δ = 0
HCT ≈ 2 RTT
Client
Server
Initial
3 packets
ACK + HS
4th
W = 1,280 B, so A(B) = 3,840 B and three packets leave.
The fourth waits until the client's Handshake (HS) packet
lifts the 3× limit.
(b) δ = 1
HCT ≈ 1 RTT
Client
Server
Initial
4 packets
W + δ = 1,281 B. One more byte raises A(B) = ⌈3B/P⌉P to
5,120 B, so all four packets leave at once.
(c) δ = 1,
Initial lost
PTO wait, 200 ms
HCT ≈ PTO + RTT
Client
Server
× Initial lost
No ACK arrives, so the client waits for PTO to
expire, then quic-go sends two probes.
2 probes
4 packets
(d) δ = 1 + parity,
Initial lost
HCT ≈ 1 RTT
Client
Server
parity
4 packets
The server rebuilds the Initial (square). Only the rebuilt
W + δ = 1,281 B counts, not the parity, so A(B) = 5,120 B.
0
50
100
150
200
250
Time since the client's first send (ms)